

HuffPost News Category Classification using DistilBERT Fine-Tuning

Usha Priya Krishnasamy

1 Introduction and Problem Definition

1.1 Problem Description and Motivation

The rapid evolution of digital journalism has created significant logistical challenges due to the overwhelming quantity of content being generated every day. As a large publisher, such as HuffPost, generated over 200,000 articles during a six-year period, accurately categorizing that content in an efficient manner (such as assigning an article to one of the major news categories) becomes a top priority. In particular, this classification issue will involve determining what category a news article belongs to (e.g. Politics, Wellness, Entertainment) based primarily on its title (the headline) and a brief description of the article.

This project is motivated by the need for automated content curation. Tagging articles manually is difficult, too time consuming, expensive and inconsistent. Therefore, if a news company could create a high accuracy classification model to automate how to route news feeds, improve search engine optimization (SEO) efforts, and provide data for creating a recommendations engine to keep users returning to the site, they would benefit from higher click-through rates (CTR) and improved ad targeting due to providing the most relevant readers with the content available.

1.2 Project Goals

Business goal: The main goal is to cut the cost of tagging content by 80% while keeping the same accuracy as a human editor. This fits into a bigger effort to deliver content to users quickly both on digital platforms and through newsletters.

Data-science goal: The technical goal is to create a robust multi-class classifier that can deal with substantial class imbalance and semantic ambiguity in the classification prediction process. Specifically, we are targeting an estimated test accuracy $> 60\%$ and a Macro-F1 > 0.55 on our refined data set. The model is expected to classify approximately 30 unique categories using only short-text inputs.

1.3 Dataset Description

The HuffPost News Category Dataset from the HuggingFace datasets library contains 200,853 total articles and 41 different categories. All input features in the article are text based. Each record contains: a short headline (~ 8 words), short_description (~ 18 words), an author field, a publication date and category (the target label). No numerical, date or image features are used in the modeling. The concatenated headline and description text with [SEP] token, with median length of ~ 29 words, serve as the sole input to the model.

Challenges of the dataset: The key challenges identified are: (1) *Class Imbalance*: The dataset has a severe class imbalance, in that Politics (32,739 articles) is by far the majority class while Education (1,004 articles) and Latino Voices have a significantly fewer number of articles. (2) *Noise and Sparsity*: About 9.8% of the articles are missing a short description, so they only have a headline. There are also 488 articles that show up more than once in the dataset. (3) *Semantic Overlap*: Multiple categories have overlapping themes; i.e. Wellness and Healthy Living or Arts and Arts & Culture. (4) *Temporal Drift*: The dataset spans 2012 to 2018, meaning the vocabulary and topic distribution may not reflect current news cycles, which could limit model performance when applied to post-2018 content.

EDA Summary: After removing duplicates, empty descriptions and merging the 8 overlapping category pairs, the cleaned dataset has 176,731 articles and 29 categories. The text length distribution shows $\text{MAX_LENGTH} = 64$ tokens covers 98.4% of articles without truncation.

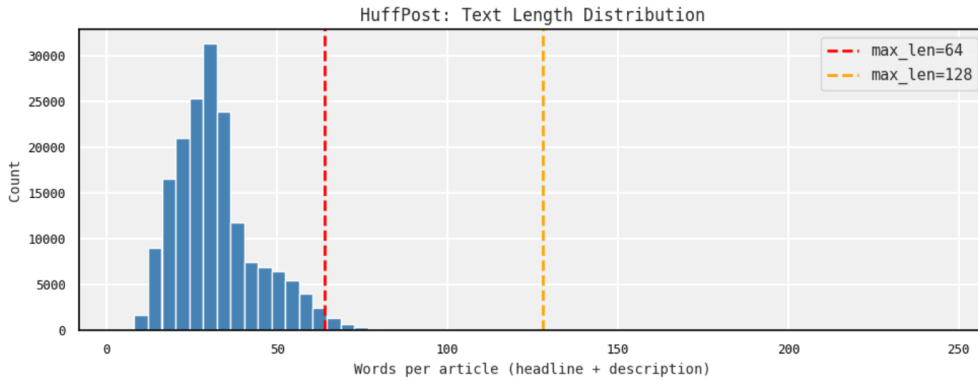


Figure 1: Text length distribution (words per article). The dashed red line at $\text{MAX_LEN} = 64$ tokens covers 98.4% of articles without truncation.

2 Methodology

2.1 Preprocessing Pipeline

Data preparation was a key factor in producing a successful model. The pre-processing pipeline contained four main stages:

- **Cleaning and Refinement:** Duplicates were removed (488 rows) and articles with null descriptions were removed. Small class counts were merged into larger and similar categories (e.g., "CULTURE ARTS" into "ARTS") and the "tiny" categories with less than 1,000 samples that could not be reasonably merged were removed. This produced a cleaned, refined set of 29 classes.
- **Text Synthesis:** The headline and short description fields were concatenated, separated by a standard [SEP] token. As a result, the model is able to receive the greatest amount of context for each article.
- **Tokenisation:** The `DistilBertTokenizerFast` was used for tokenization. Text was truncated or padded to a maximum length of 64 tokens. This ensured consistent input size while balancing efficiency and information retention. Analysis showed that 98.4% of articles were within the maximum length, so very little information was lost.
- **Stratified Split:** The cleaned data were divided 70 / 15 / 15: training (123,711), validation (26,509), and test (26,511), to preserve each class's proportion across 3 splits and prevent data leakage. The test set was held out completely and evaluated exactly once after the best model weights were restored, ensuring no leakage into model selection

2.2 Model Architecture and Justification

The final architecture is a fine-tune DistilBERT (distilbert-base-uncased) because DistilBERT retains 97% of BERT's performance while being 40% smaller than BERT and 60% faster than BERT. During Milestone 2, three models were contrasted to evaluate the effectiveness of fine-tuning to achieve desired results from this dataset. A baseline model, a frozen DistilBERT model and a fully

fine-tuned DistilBERT model were evaluated. The fully fine-tuned DistilBERT model produced superior results due to the model's weights being adapted to the unique "journalistic" vocabulary of news headlines. The architecture used for the model consists of a transformer backbone followed by a dropout layer ($p=0.3$) and linear classification head. The model was trained using PyTorch with mixed precision training via `torch.amp.autocast` and GradScaler, reducing GPU memory consumption and accelerating training without sacrificing numerical precision.

Training strategy: A class-weighted cross-entropy loss function with label smoothing (0.1) was implemented to address class imbalance and enhance generalization. The class weighting penalized errors on underrepresented classes more heavily, therefore reducing model bias toward the majority class. Label smoothing reduced overconfidence in predictions and helped improved generalization to unseen data. The model was optimized with the AdamW optimizer (learning rate = 2×10^{-5} , weight decay = 0.01). AdamW works well for transformer models because it separates weight decay from gradient updates.

A small learning rate was used to fine-tune the pretrained DistilBERT weights carefully, so the original language representations were not lost. Gradient clipping `max_norm = 1.0` was used to stop gradients from becoming too large during fine-tuning of deep transformer models. This helped keep training stable and made convergence smoother. Overall, the training strategy was designed to balance three key objectives: handling class imbalance, preserving pretrained knowledge, and achieving stable and efficient convergence.

Callback: To train the model, a few techniques were used to help it learn better. First, we used a warm-up scheduler, which slowly increases the learning rate at the start so the model doesn't change too fast and lose what it already learned. We also used early stopping, which automatically stops training when the model stops improving. It checks the validation score after each round, and if there's no improvement after 2 rounds in a row, it stops. In our case, training stopped at round 14, and the best version of the model (from round 12, with a score of 0.6637) was saved and used for the final test. The test set was never exposed during training or hyperparameter tuning; it was evaluated exactly once using the restored best-epoch weights, ensuring no data leakage into model selection.

3 Results and Evaluation

The model did well overall, especially given the fact that it had to choose between 29 different categories using short pieces of text.

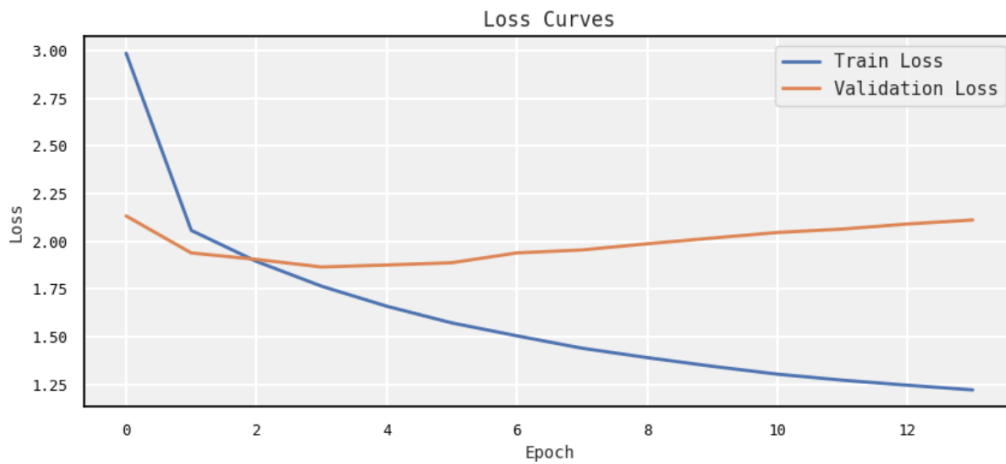


Figure 2: Training vs. validation loss curves over training epochs.

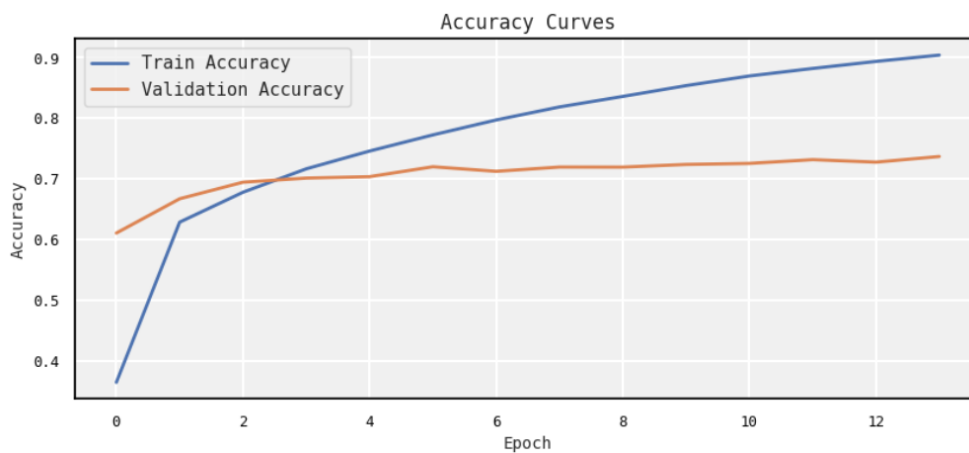


Figure 3: Training vs. validation accuracy curves over training epochs.

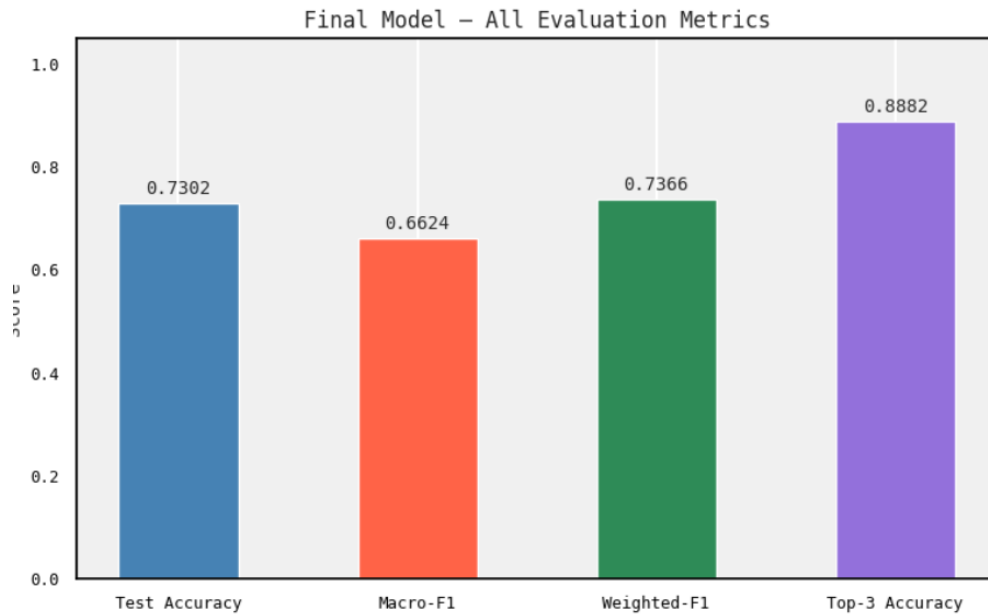


Figure 4: Final model evaluation metrics on the test set: accuracy, Macro-F1, Weighted-F1, and Top-3 accuracy.

The training accuracy (88.22%) was noticeably higher than the test accuracy (73.02%), which means the model memorized the training data a bit too well; this is pretty common when fine-tuning Transformer models on small, specific datasets. Despite this, the validation and test scores were almost identical (73.18% vs 73.02%), which is a good sign that the model still works well on data it hasn't seen before. The F1 score of 0.66 also beat our target of 0.55, which is a solid result for a 29-class problem where some categories had much less data than others.

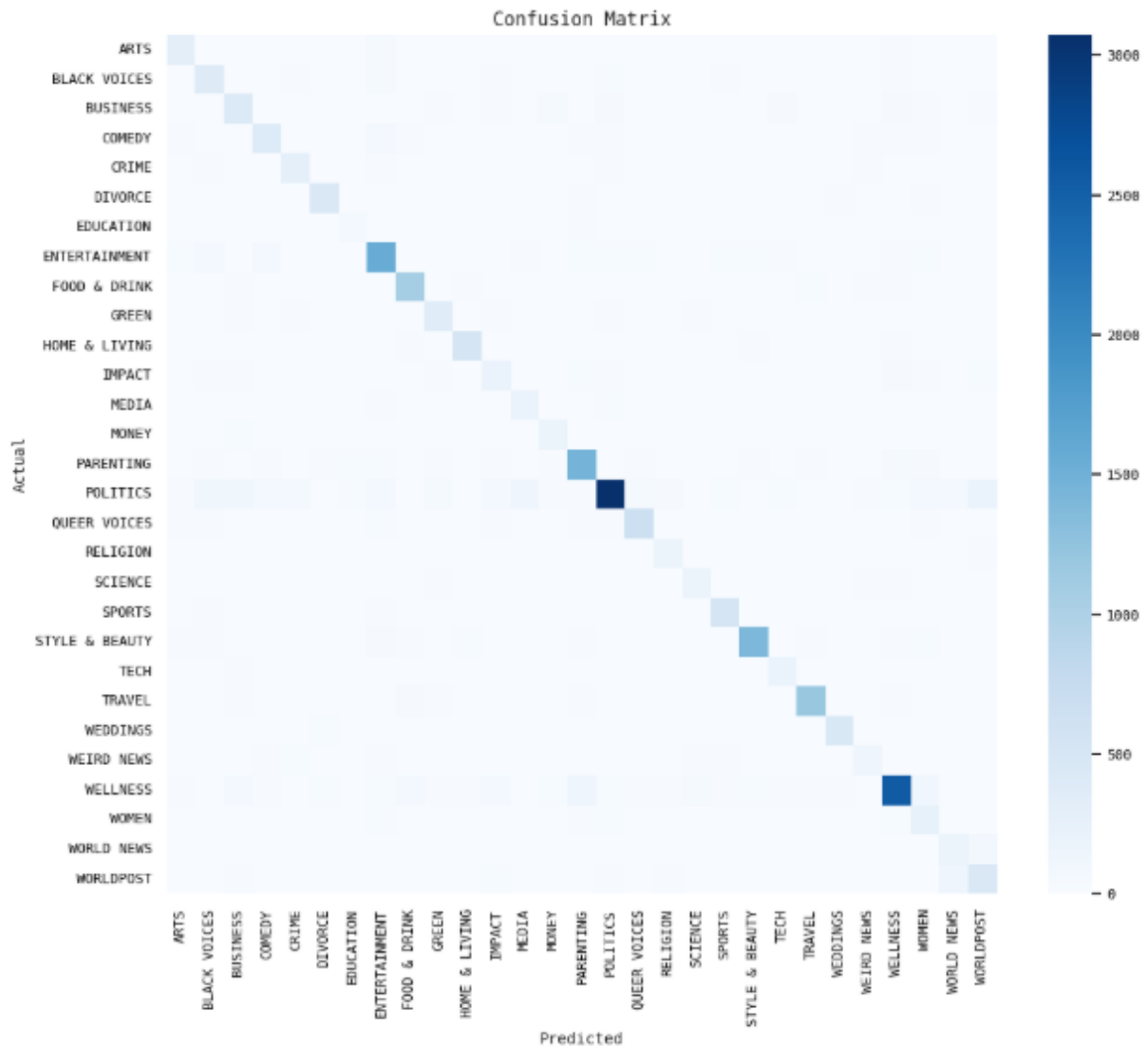


Figure 5: Confusion matrix on the held-out test set (29 classes).

Looking at the confusion matrix, categories with clear vocabulary like STYLE & BEAUTY, TRAVEL, and FOOD & DRINK had the highest accuracy. POLITICS and ENTERTAINMENT also did well because of their unique words and larger sample sizes. However, the model struggled to tell apart similar categories such as WORLDPOST and WORLD NEWS, or COMEDY and ENTERTAINMENT. Since the input text is only about 29 words on average, there often isn't enough difference between these classes to make a clear prediction. We consider this model our best option because it balances speed and accuracy. Although a larger model like RoBERTa-Large might improve accuracy, the small gain does not justify the much higher computational cost for

real-time news use.

Table 1 shows the top-5 and bottom-5 categories by F1-score; the full per-class breakdown is provided in the Appendix Table 7

Table 1: Top-5 and bottom-5 categories by F1-score on the held-out test set.

Category	Precision	Recall	F1-Score	Support
<i>Top 5 (strongest performance)</i>				
STYLE & BEAUTY	0.887	0.844	0.865	1660
TRAVEL	0.881	0.818	0.848	1410
FOOD & DRINK	0.822	0.867	0.844	1224
WEDDINGS	0.807	0.870	0.837	547
DIVORCE	0.816	0.839	0.827	514
<i>Bottom 5 (weakest performance)</i>				
WORLD NEWS	0.376	0.601	0.463	326
WOMEN	0.334	0.583	0.425	458
IMPACT	0.367	0.501	0.424	459
WEIRD NEWS	0.462	0.527	0.492	332
MEDIA	0.383	0.713	0.498	341
Macro avg	0.632	0.700	0.658	26,510
Weighted avg	0.754	0.721	0.730	26,510

4 Discussion and Reflection

4.1 Limitations and Sources of Error

The most significant factors limiting the performance of this model are caused by the small number of words that are present in all inputs. Many news outlets use catchy headlines to draw the reader in, similar to click baiting, which doesn't provide a lot of semantic information for a classifier to work with. The extreme class imbalance in this dataset means that when a classifier

is uncertain about what class of output to give, it will select the most frequent class. Lastly, this dataset covers the period from 2012 to 2018 and the topics and vocabularies of the news cycle may not be current and thus this may limit the performance of the model for use in news classification tasks in 2026 unless further trained on new data. Beyond the data, the model itself has inherent limitations. DistilBERT's attention mechanism, while powerful, is sensitive to the choice of learning rate during fine-tuning, too high and pretrained representations are destroyed, too low and the model under-adapts. Additionally, with only ~ 29 words of input per article, the model has limited representational capacity to resolve ambiguous categories, regardless of architecture size.

4.2 Potential Improvements

Focal Loss could further penalize minority class errors beyond the class-weighted cross-entropy already applied; this may lead to an increase in the Macro-F1 score. Incorporating auxiliary inputs such as "author" or "date" could assist the model in learning about how different news categories change over time. Using back-translation (for example, translating English to French and back to English again) for data augmentation could generate synthetic training data for low-volume categories such as EDUCATION.

4.3 Reflection on What We Learned

Overall, cleaning up the data made a bigger difference than changing the model itself. When we combined some categories and removed messy data, our results improved more than when we adjusted the model's settings. We also learned that starting with a pre-trained model (DistilBERT) was really helpful, it gave us a decent starting point right away, and we were able to build on that from there. We also noticed that how we set up the training mattered a lot, warming up the learning rate slowly at the start meant the model didn't lose what it had already learned, and stopping early saved us from overfitting. If we did this project again, we would spend more time figuring out which categories the model keeps mixing up and focusing on those specifically. We would also think more carefully upfront about which categories are too similar to tell apart from a short headline alone, and either merge them or collect better data before training.

4.4 Conclusion

The short input text is a significant limitation since most news headlines are made to have impact instead of describing something which doesn't provide enough meaning for a classifier to use as an indicator. Another limitation of the data is that there is a large class imbalance where the model typically will default towards high frequency classes when it is uncertain. The dataset covers 2012 to 2018, so performance may degrade on more recent news without retraining. Future improvements include applying Focal loss for minority classes, adding author or date as auxiliary inputs, and back-translation to generate synthetic data for small categories like EDUCATION. The key lesson from the project was that data cleaning decisions, merging overlapping categories and removing noise improved performance rather than hyperparameter tuning.

Overall, this project showed how useful transfer learning can be. Our basic embedding model started at 53% accuracy and once we fully fine-tuned DistilBERT we got up to 63.99% in Milestone 2. After cleaning up the data, adding regularization, and training longer, we pushed that up to 73.02% in the final version. If we did this again, we would spend more time on hard negative mining; basically, forcing the model to practice on the examples it keeps getting wrong so that it can tell similar categories apart better.

Appendix: Additional Visualisations and tables

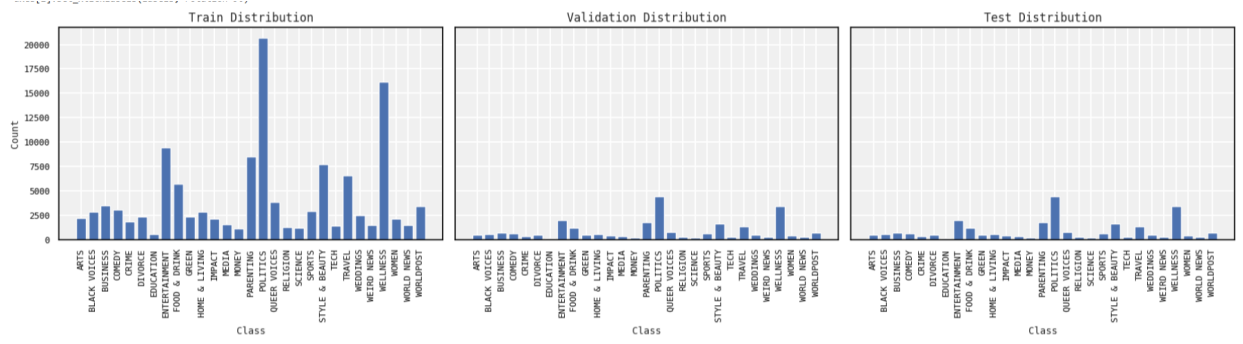


Figure 6: Class distribution across train, validation, and test splits, confirming stratified sampling preserved class proportions.

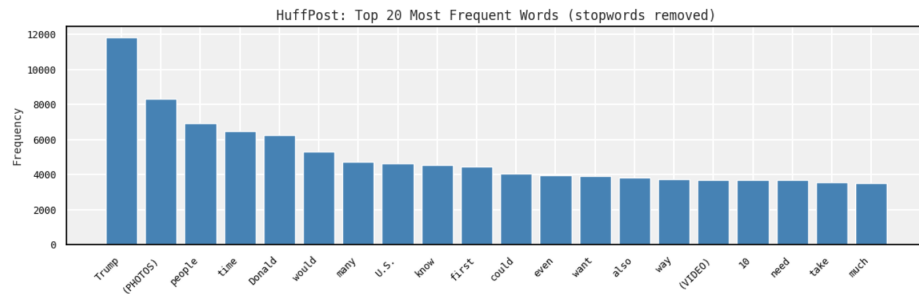


Figure 7: Top 20 most frequent words after stopwords removal. Dominant terms reflect the political and lifestyle focus of HuffPost content.

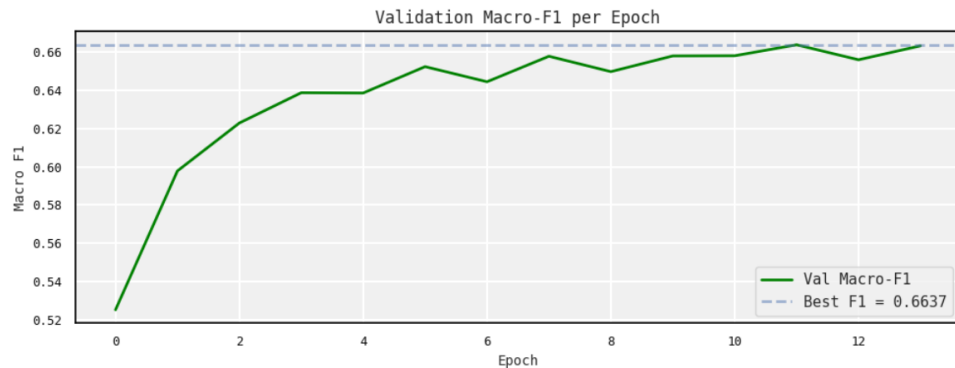


Figure 8: Validation Macro-F1 score per epoch. Best F1 = 0.6637 achieved at epoch 12; early stopping triggered at epoch 14.

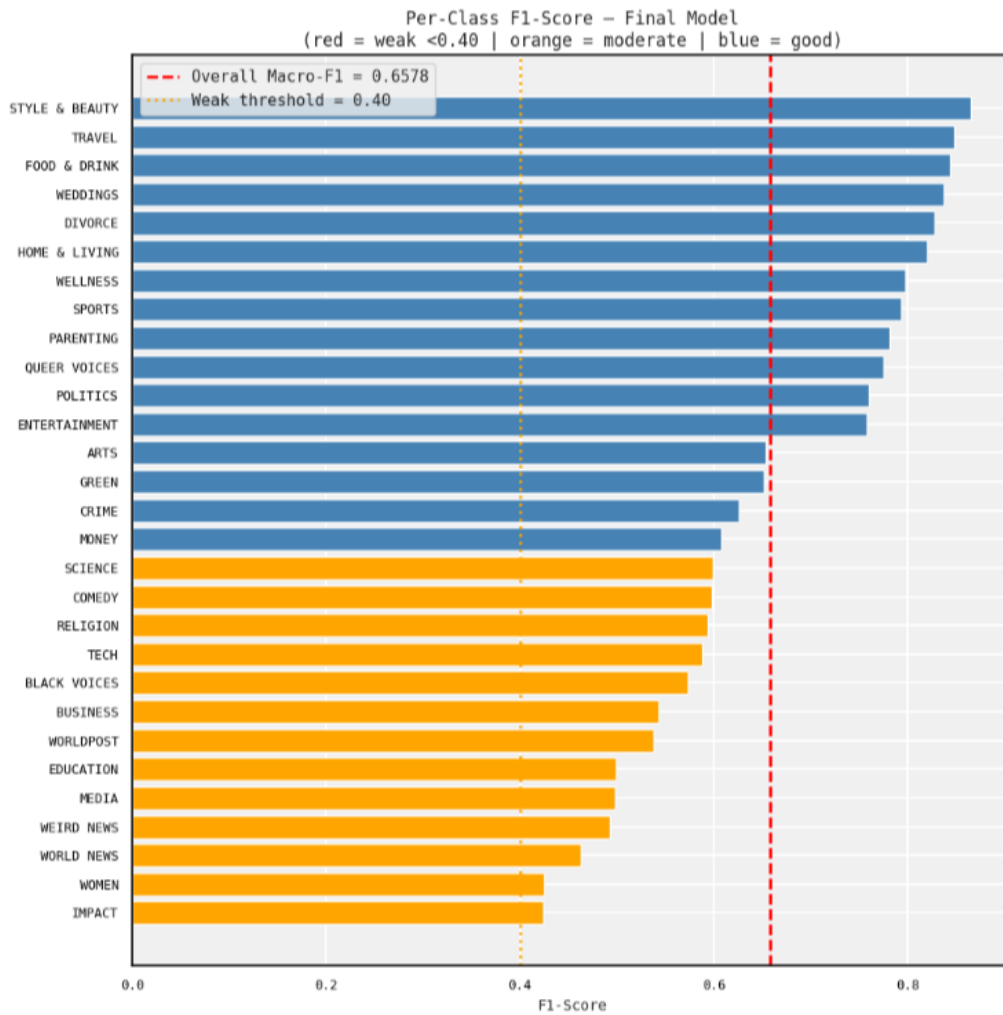


Figure 9: Per-class F1-score for all 29 categories. Red bars indicate weak classes ($F1 < 0.40$), orange moderate, blue strong (≥ 0.60).

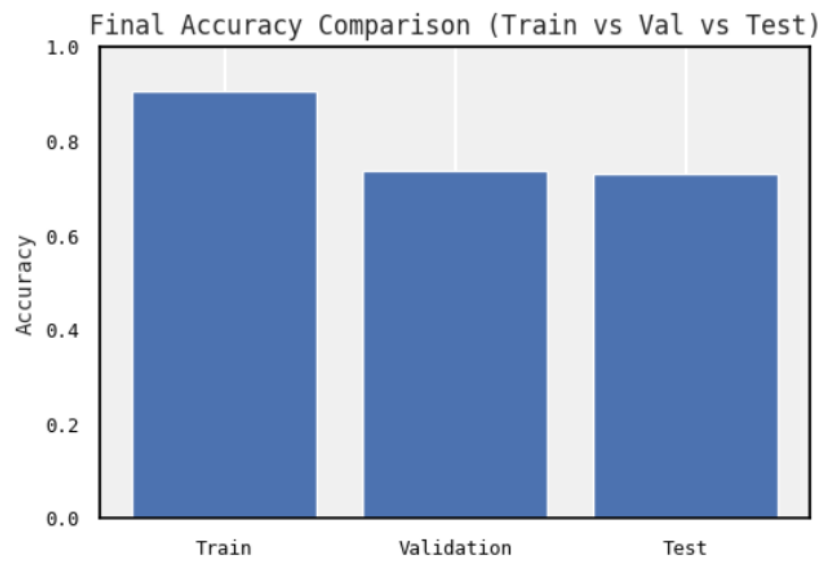


Figure 10: Final accuracy comparison across train, validation, and test splits.

Tables

Table 2: Category merge map applied to reduce ambiguity (Table 1).

Original Label	Merged Into
HEALTHY LIVING	WELLNESS
ARTS & CULTURE	ARTS
CULTURE & ARTS	ARTS
TASTE	FOOD & DRINK
THE WORLDPOST	WORLDPOST
PARENTS	PARENTING
ENVIRONMENT	GREEN
STYLE	STYLE & BEAUTY

Table 3: Preprocessing cleaning summary (Table 2).

Step	Rows Removed	Reason
Exact duplicates	488	Prevent data leakage
Missing descriptions	19,611	Insufficient text for classification
Category merging	—	Reduced ambiguity (41 $\rightarrow$ 33)
Tiny categories dropped	4,023	Too few samples; now 29 categories
Total removed	24,122	
Final dataset	176,731 articles, 29 categories	

Table 4: Milestone 2 model comparison — basis for final model selection (Table 3).

Model	Test Acc	Macro F1	Train Time
Baseline (Embedding + GAP)	0.5331	0.4557	$\approx$ 40 s
Custom (GAP + BatchNorm)	0.4793	0.4220	$\approx$ 2 min
DistilBERT Frozen	0.5805	0.5117	$\approx$ 2 hr 5 min
DistilBERT Full FT	0.6399	0.5796	$\approx$44 min

Table 5: Final model hyperparameters.

Parameter	Value
Learning rate	2×10^{-5}
Batch size	32
Max epochs	25
Dropout rate	0.3
Weight decay	0.01
Max token length	64
Num classes	29
Random seed	42

Table 6: Final model performance summary on held-out test set (Table 4).

Metric	Score
Test Accuracy	0.7302
Macro F1	0.6624
Weighted F1	≈ 0.74
Top-3 Accuracy	0.8882
Best Validation F1	0.6637 (epoch 12)
Best Validation Acc	0.7318
Training Time	≈ 1 hr 44 min
Best Epoch	12 / 25 (early stop at epoch 14)

Table 7: Per-class precision, recall, and F1-score on the held-out test set (29 categories), sorted by F1-score descending.

Category	Precision	Recall	F1-Score	Support
STYLE & BEAUTY	0.887	0.844	0.865	1660
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WEDDINGS	0.807	0.870	0.837	547
DIVORCE	0.816	0.839	0.827	514
HOME & LIVING	0.789	0.853	0.820	624
WELLNESS	0.870	0.737	0.798	3462
SPORTS	0.763	0.825	0.793	633
PARENTING	0.761	0.802	0.781	1824
QUEER VOICES	0.775	0.775	0.775	835
POLITICS	0.918	0.649	0.760	4435
ENTERTAINMENT	0.789	0.730	0.758	2020
ARTS	0.640	0.669	0.654	483
GREEN	0.614	0.693	0.651	505
CRIME	0.578	0.681	0.625	401
MONEY	0.549	0.680	0.607	256
SCIENCE	0.524	0.699	0.599	266
COMEDY	0.594	0.602	0.598	663
RELIGION	0.530	0.674	0.593	279
TECH	0.501	0.711	0.588	305
BLACK VOICES	0.526	0.630	0.573	617
BUSINESS	0.496	0.599	0.543	761
WORLDPOST	0.570	0.510	0.538	736
EDUCATION	0.411	0.634	0.499	134
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